Low Cost Real-Time Simulator

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***Abstract*—This paper presents a low-cost, real-time Hardware-in-the-Loop (HIL) simulation framework designed as an accessible alternative to expensive commercial testing ecosystems. The platform features a custom-designed open-architecture interface card that handles high-fidelity signal conditioning and bidirectional level shifting between a control microcontroller and the plant model and another plant card serving as the Plant system. Two real-time emulation strategies for a power electronic converter are validated: a discrete-time averaged model executing on a C2000 digital signal processor (DSP) and a high-frequency switching model deployed on a field-programmable gate array (FPGA). Experimental hardware waveforms demonstrate precise tracking of both steady-state dynamics and transient oscillations, proving that high-diagnostic accuracy can be achieved at a fraction of commercial costs.**

***Index Terms*—component, formatting, style, styling, insert (choose 3–5 keywords)**

I. Introduction

There are two parts to this project :

1) Hardware Interface Layer: A custom, low-noise interface PCB that handles the critical analog voltage scaling, protection clamping, and digital level shifting needed to seamlessly couple a target microcontroller with the simulation environment, and a Plant PCB to serve as Host
2) Plant Emulation Layer: A dual-platform processing approach featuring a computationally efficient averaged model running on a digital signal processor (DSP) alongside a high-fidelity switching model deployed on a resource-constrained field-programmable gate array (FPGA).

II. Hardware-in-Loop Interface Card

A. Motivation and Overview

A commercial Hardware-in-Loop (HIL) simulator such as the Typhoon HIL404 requires a dedicated interface card to bridge the simulator's analog and digital I/O to the microcontroller under test. The proprietary TI LaunchPad interface card offered by Typhoon HIL is available at a significant cost (approximately 1,00,000), yet its core functionality—signal conditioning, level shifting, and connector routing—can be replicated with a custom PCB at a fraction of the cost. This work presents a custom-designed HIL interface card built specifically to (i) replace the commercial Typhoon TI interface card for HIL experiments, and (ii) serve as a generic interface that connects to a custom plant card for fully in-house HIL testing.

The Typhoon HIL404 exposes its I/O through two DIN41612 Type-C Harting connectors (96-pin, male) on its front panel, labeled *Analog I/O* and *Digital I/O*, along with CAN, GPIO, and Ethernet ports [6]. The designed interface card mates directly to these connectors via right-angle female DIN41612 connectors mounted on the PCB. The control DSP is the TMS320F28379D, seated directly on the interface card via BoosterPack headers on a TI F28379D LaunchPad.

B. I/O Requirements

For closed-loop boost converter control, the interface card was designed to support the channels summarised in Table I. The HIL outputs simulated plant quantities (inductor current, output voltage) as analog signals to be read by the DSP ADC, while the DSP PWM outputs feed back as digital switching commands to the HIL.

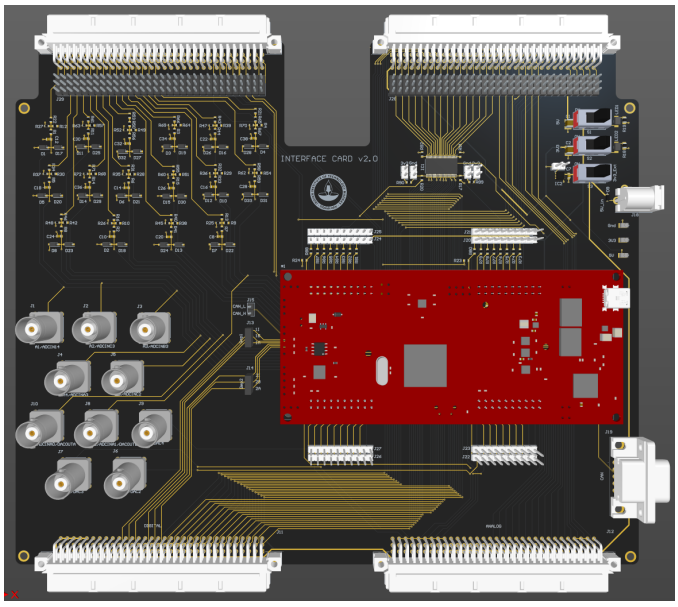


Fig. 1. 3D render of the HIL Interface Card (v1.0). BNC connectors for oscilloscope monitoring are visible on the lower left; DIN41612 Harting connectors for Typhoon HIL mating are at the top.

C. Analog Signal Conditioning

The Typhoon HIL DAC outputs span $\pm 10\,\mathrm{V}$, while the TMS320F28379D ADC accepts only 0–3 V. Each of the 16

TABLE I
INTERFACE CARD I/O SUMMARY

Signal	Count	Notes
HIL Analog Out (AO)	16	Scaled and clamped to 0–3 V
Direct ADC In (AI)	6	Plant card or external analog
HIL Digital In (DI)	16	DSP PWM/GPIO
HIL Digital Out (DO)	16	Shifted from 5 V to 3.3 V
eQEP channels	2	Quadrature encoder inputs
CAN	1	DB9 (CAN_P, CAN_N)

analog channels therefore employs a resistor divider ($R_{\text{top}} = 100\,\text{k}\Omega$, $R_{\text{bot}} = 33.2\,\text{k}\Omega$) biased from the 3.3 V rail to map the ± 10 V range to 0–3 V at the ADC pin. A precision 3.83 kΩ series resistor (ERA3AEB3831V) limits input current, and a Schottky diode clamp ensures the ADC pin cannot exceed 3.3 V or fall below 0 V under any HIL output condition. Decoupling capacitors at each ADC input suppress high-frequency switching noise. Two channels (AO8, AO16) are shared with the DSP DAC outputs (ADCINA0/DACA, ADCINA1/DACB) via zero-ohm jumpers, allowing the same BNC connector to serve as either input or output.

D. Digital Level Shifting

The HIL digital outputs are 5 V LVCMOS; the DSP operates at 3.3 V. The SN74LVC16T245 bidirectional 16-bit bus transceiver (IC1) handles all 16 channels with separate supply rails (VCCA = 3.3 V, VCCB = 5 V). A critical issue was identified during bring-up: the OE pin had been hard-tied to GND, causing the transceiver to drive the bus before the DSP output buffers had initialised, resulting in erroneous logic levels and potential latch-up. The fix routes OE through a pull-up network (R17, R20), ensuring high-Z during power-up and correct sequencing thereafter. This was validated on a revised PCB across multiple power cycles.

E. Analog Monitoring and Plant Card Interface

Eight BNC connectors (J1–J10) provide direct oscilloscope access to selected analog channels—inductor current, output voltage, and DAC references—without additional probing fixtures. For plant card interfacing, connectors J28 and J29 (TSW-132-07-G-T, 96-pin) mirror the DIN41612 pinout, allowing the plant card to substitute the HIL in standalone DSP-to-DSP testing. Since both boards operate at 3.3 V, no level shifting is required on this path. Decoupling capacitors (100 nF, 0603) are placed at each ADC input to suppress inter-board coupling.

F. Power Supply

Supply rails are sourced from the HIL connector or the LaunchPad board via jumper selection. An on-board LDO (TPS78533, IC2) provides a regulated 3.3 V for the analog front end. Status LEDs (LED1, LED2) with current-limiting resistors indicate the presence of the 5 V and 3.3 V rails for rapid power-on verification.

G. Testing and Verification

Supply rails were confirmed within $\pm 1\%$ of nominal. All 16 analog scaling channels were characterised by applying a known input voltage and measuring the ADC pin voltage; all channels were within the expected range. Mechanical fit against the Typhoon HIL404 chassis was confirmed, and level-shifter logic levels were verified by toggling DSP GPIO pins and probing both sides of IC1. The OE sequencing fix was confirmed functional on the revised board. as subsequently fabricated and validated.

III. PLANT CIRCUIT BOARD

The plant PCB serves as the real-time plant emulator in the low-cost HIL setup. Rather than using an FPGA-based high-frequency switching model (as in commercial HIL systems such as the Typhoon HIL404), the initial implementation runs an averaged boost converter model on a TMS320F28379D DSP, providing a cost-effective first step toward real-time simulation.

The board is a carrier PCB for the F28379D LaunchPad, which sits directly via BoosterPack headers. Two DIN41612 Type-C connectors (J1, J2) on the top edge interface directly with the HIL interface card, using the same 96-pin pinout as the Typhoon HIL front-panel connectors. This allows the plant PCB to be substituted in place of the Typhoon HIL during standalone DSP-to-DSP testing. The connector pinout on the plant card mirrors that of the interface card, so analog outputs from the plant DSP (DAC channels, PWM-based DAC signals) are mapped to the corresponding analog input channels on the interface card without additional rewiring.

Since both the plant card and interface card operate at 3.3 V logic, no level shifting is required on this interface. Decoupling capacitors (C1–C5) are placed at the ADC input pins to suppress high-frequency noise. Small-signal connectors (J3, J4) provide 5 V and 3.3 V test points for power rail monitoring during bring-up. The LaunchPad's USB port remains accessible for code flashing and serial debugging.

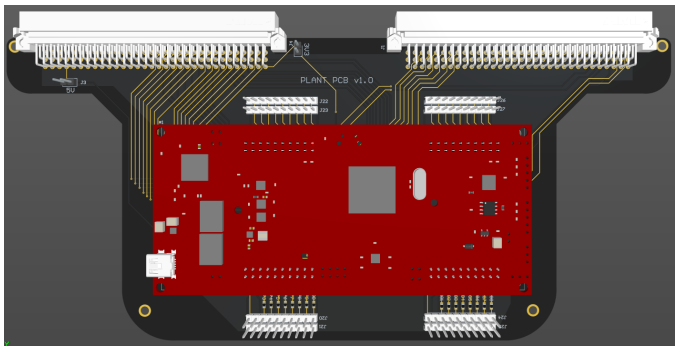


Fig. 2. Render of Plant PCB (v1.0)

IV. AVERAGED CONVERTER MODEL IN C2000 DSP

The averaged model describes the boost converter dynamics through two first-order ODEs. For duty cycle D and drive enable signal D_{en}, the effective duty cycle is $D_{\text{eff}} = D \cdot D_{\text{en}}$. The state equations are:

$$V_L = V_{\text{in}} - (1 - D_{\text{eff}})\, V_C \tag{1}$$

$$I_C = (1 - D_{\text{eff}})\, I_L - I_o, \quad I_o = \frac{V_C}{R} \tag{2}$$

The states I_L and V_C are obtained by integrating V_L/L and I_C/C respectively. The Tustin (bilinear) method is used for discrete-time integration:

$$I_L[n] = I_L[n{-}1] + \frac{T_s}{2L}\big(V_L[n] + V_L[n{-}1]\big) \tag{3}$$

$$V_C[n] = V_C[n{-}1] + \frac{T_s}{2C}\big(I_C[n] + I_C[n{-}1]\big) \tag{4}$$

where $T_s = 1/f_{\text{sim}}$ is the simulation timestep. The Tustin method is preferred over forward/backward Euler as it preserves second-order stability and avoids artificial damping at high frequencies relative to T_s.

A. Averaged Converter Waveforms

The inductor current I_L and capacitor voltage V_C are scaled to 12-bit DAC codes as:

$$d_{I_L} = I_L \cdot \frac{2047}{5}, \qquad d_{V_C} = V_C \cdot \frac{2047}{650} \tag{5}$$

where $d \in [0, 4095]$. The oscilloscope voltage for each channel is recovered as $V_{\text{osc}} = (d/4095) \cdot V_{\text{ref}}$.

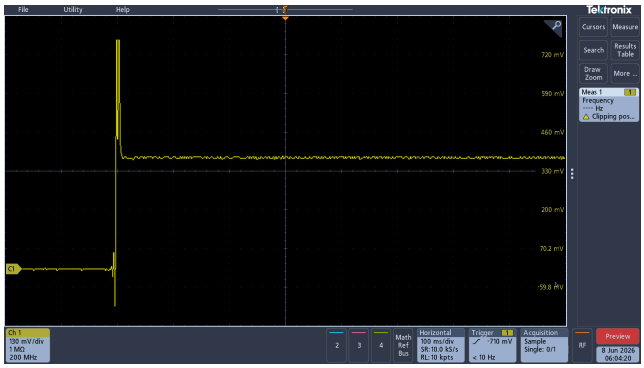


Fig. 3. Inductor Current (d = 0.334)

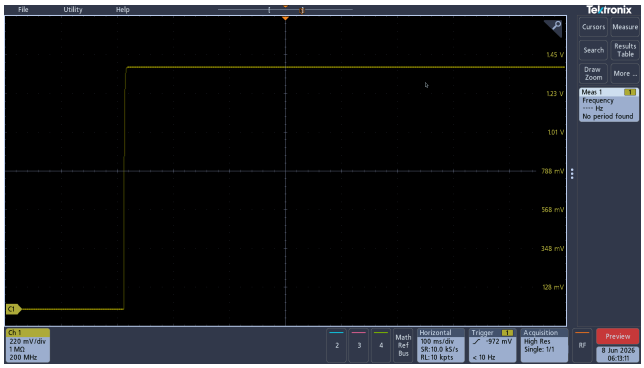


Fig. 4. Capacitor Voltage (d=0.334)

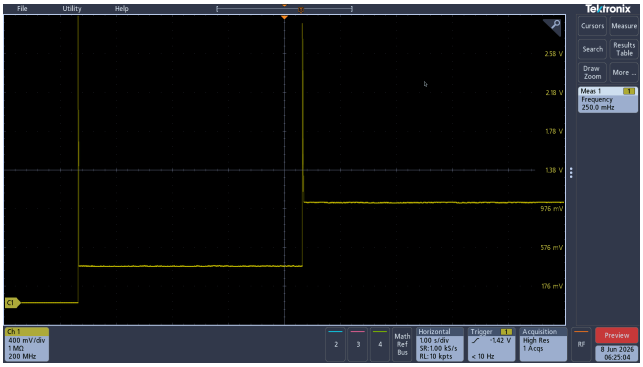


Fig. 5. Inductor Current (d: 0.334 → 0.6)

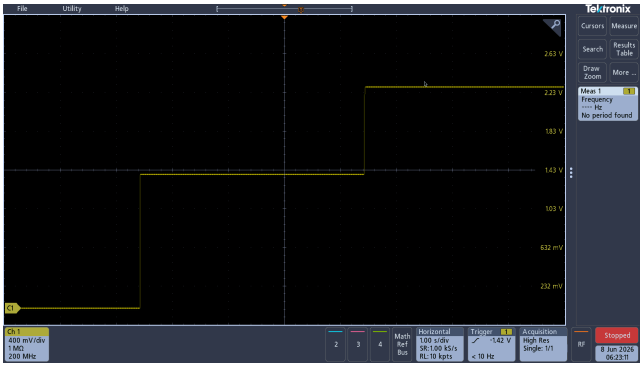


Fig. 6. Capacitor Voltage (d: 0.334 → 0.6)

V. Switching Converter Model in Altera Max 5 FPGA

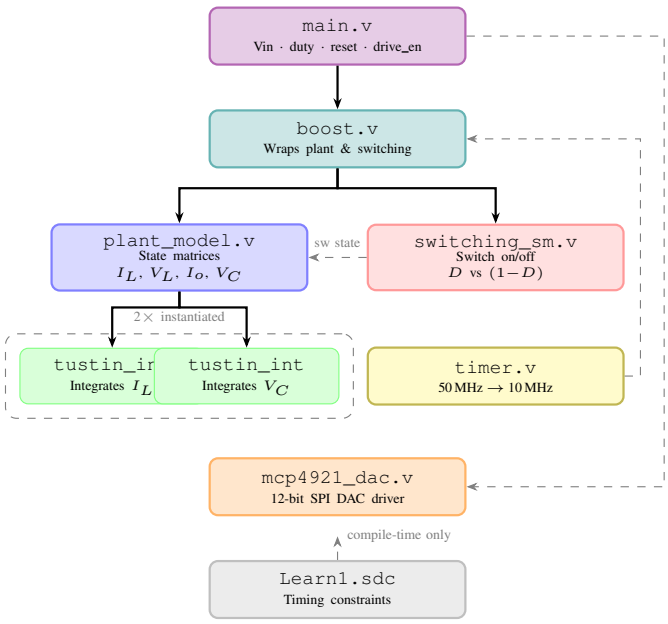


Fig. 7. Verilog Code flow diagram

A. Netlist Diagrams

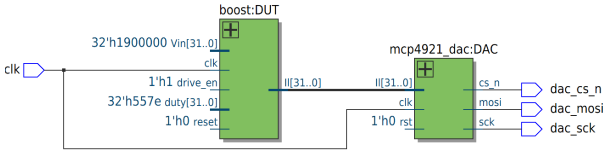


Fig. 8. Main Netlist Diagram

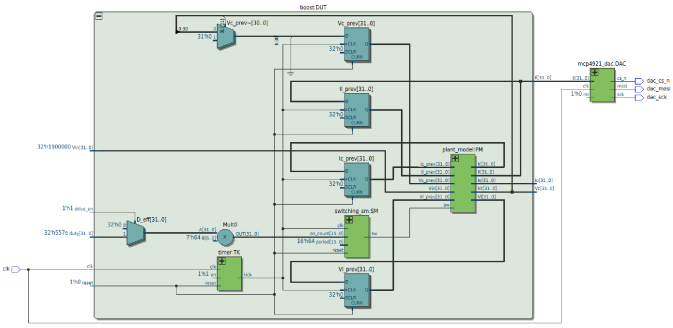


Fig. 9. Boost Netlist Diagram

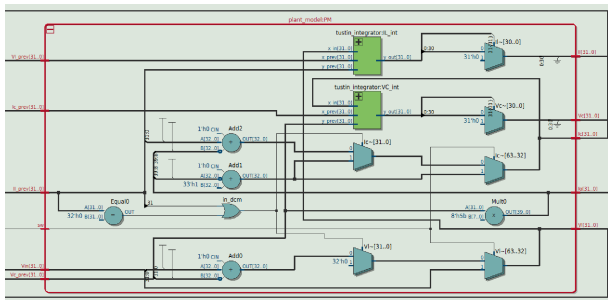


Fig. 10. Plant Model Netlist Diagram

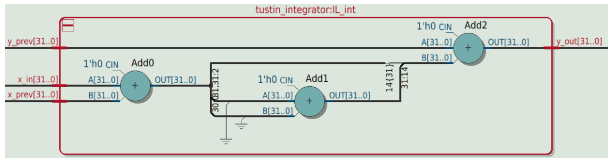


Fig. 11. Tustin Integrator Netlist Diagram

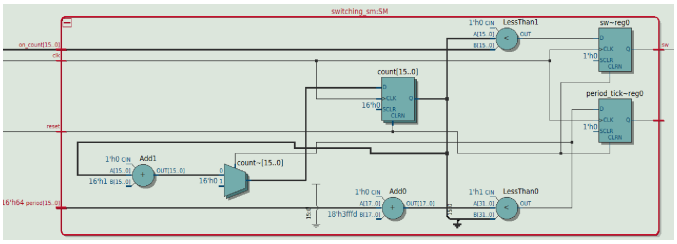


Fig. 12. Switching Netlist Diagram

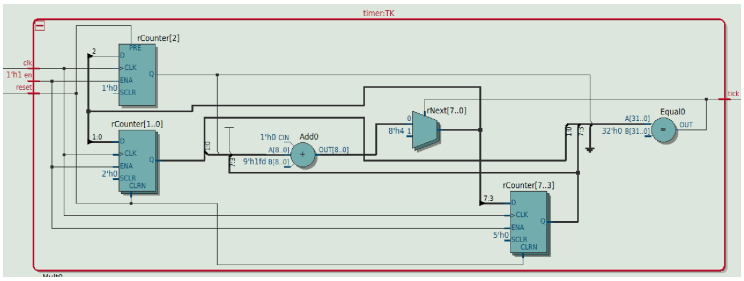


Fig. 13. Timer Netlist Diagram

VI. PLECS SIMULATION WAVEFORMS

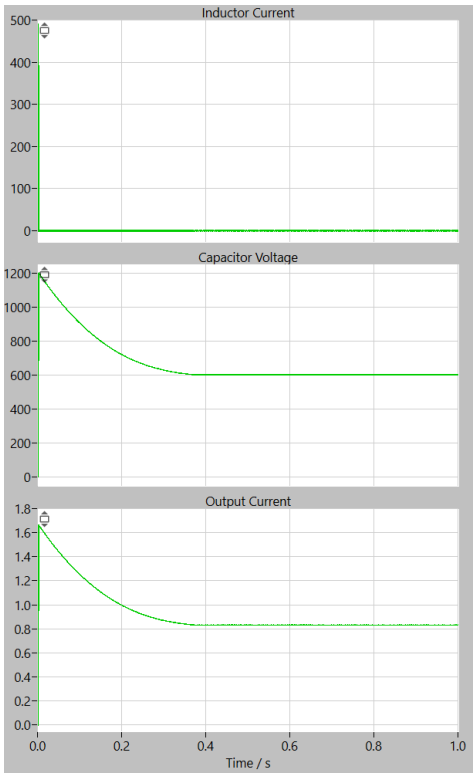


Fig. 14. Full waveform - PLECS

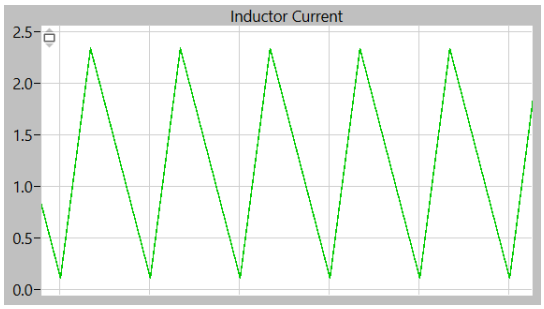


Fig. 15. Inductor Current- PLECS

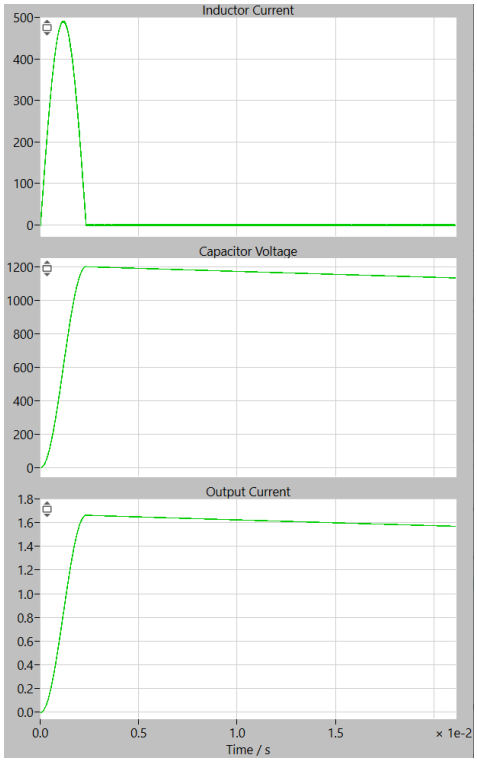


Fig. 16. Transient zoomed - PLECS

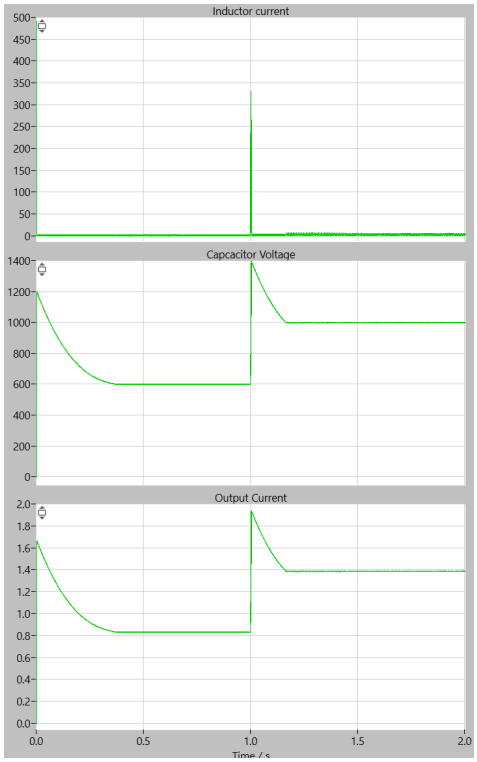


Fig. 17. Duty variation from 0.334 to 0.6

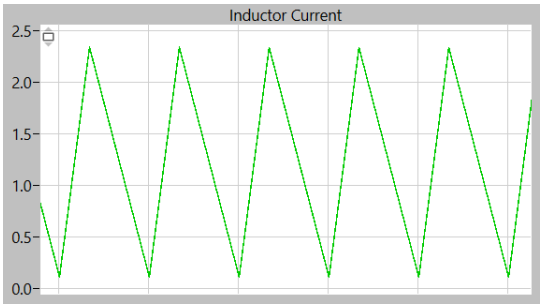


Fig. 18. Inductor Current- PLECS

VII. ModelSim waveforms

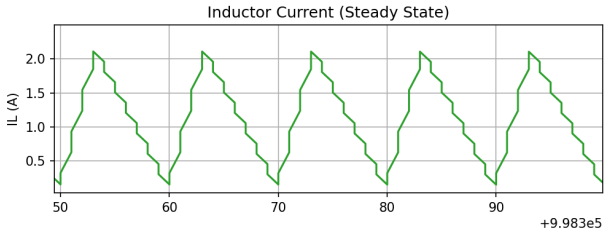


Fig. 19. Inductor Current Ripple (d=0.334)

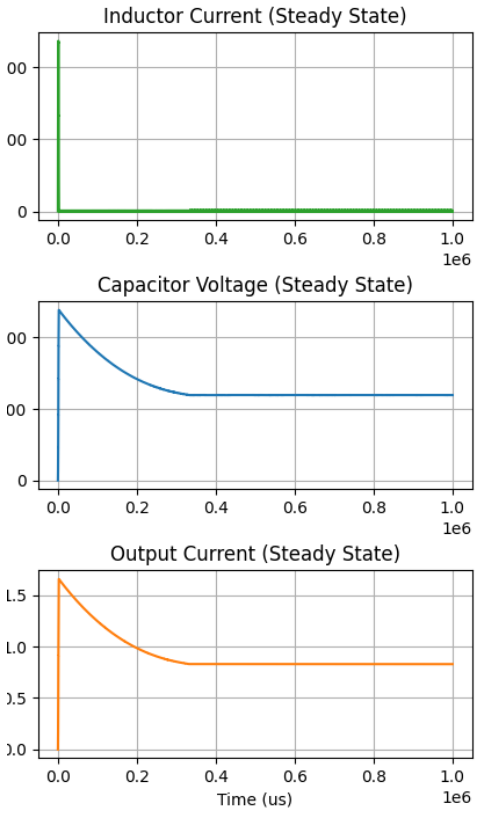


Fig. 20. Full waveform (d=0.334)

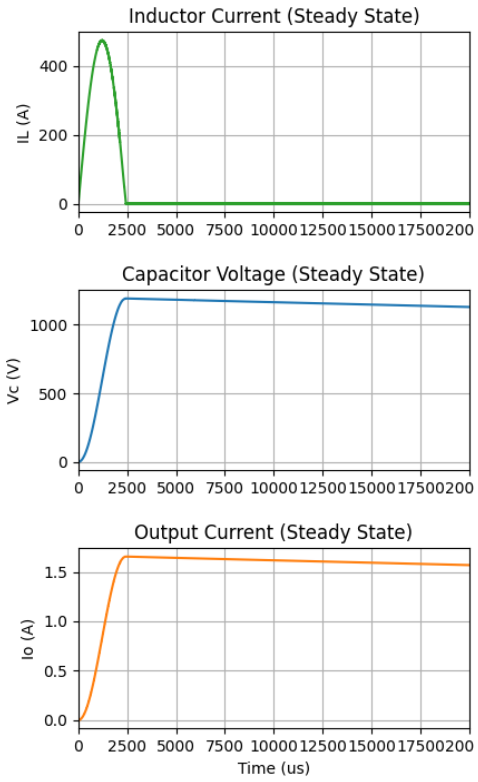


Fig. 21. Tranient zoomed waveform (d=0.334)

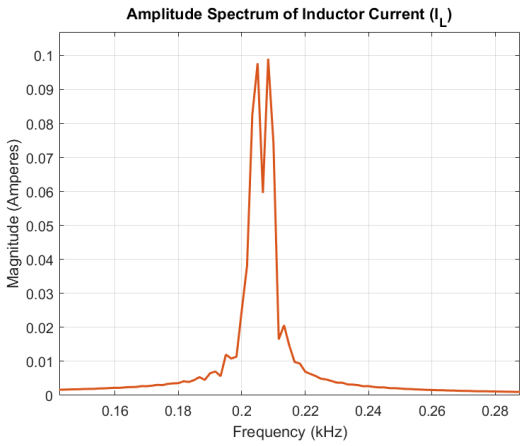


Fig. 22. Frequency Analysis of iL (d=0.334)

From the physics, we know that the Low frequency fundamental harmonic is at $\frac{1-D}{\sqrt{LC}}$ which is equal to **216.4 Hz**, from the above simulation frequency spectrum, we can also observe the low frequency ripple at **206 Hz**, which is very close to expected frequency.

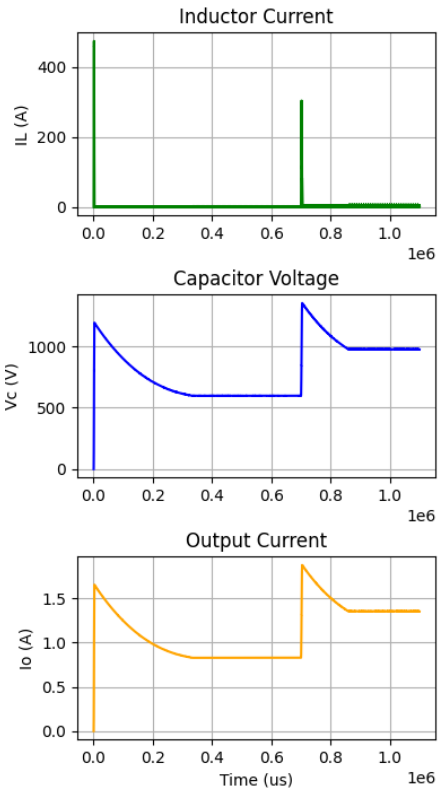


Fig. 23. Full waveform (d: 0.334 $\rightarrow$ 0.6)

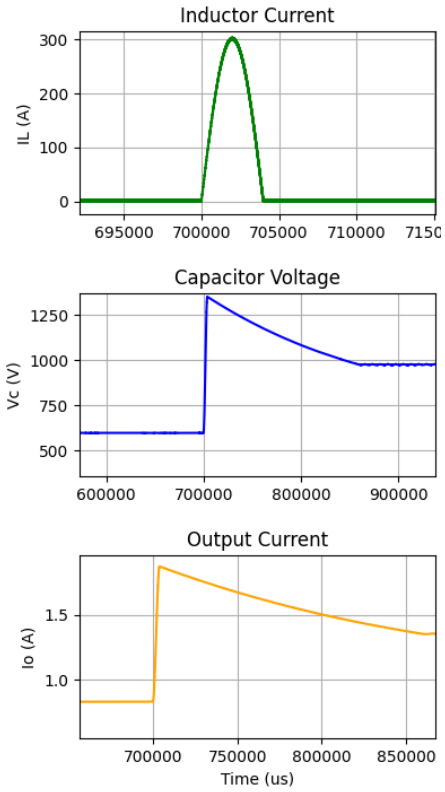


Fig. 24. Tranient zoomed waveform (d: 0.334 $\rightarrow$ 0.6)

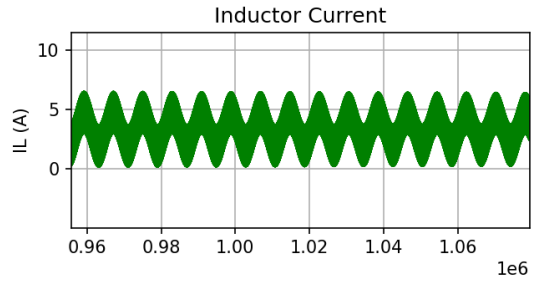


Fig. 25. Low frequency component (d = 0.6)

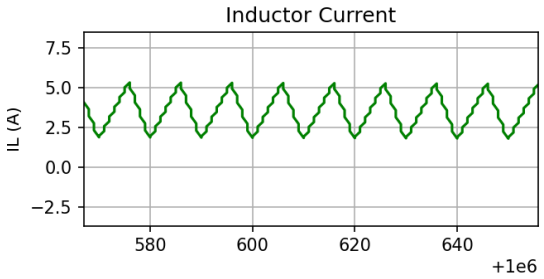


Fig. 26. Triangular inductor current (d = 0.6)

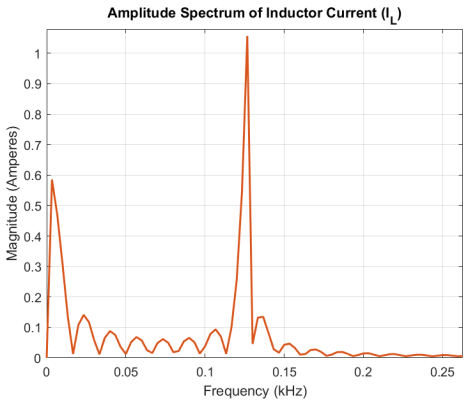


Fig. 27. Frequency Analysis of iL (d=0.6)

Low frequency component for d=0.6 is at **130 Hz**, from the above simulation frequency spectrum, we can also observe the low frequency ripple at **127 Hz**, which is very close to expected frequency.

A. Implementation in Hardware

To implement the Switching boost converter model in FPGA, I used an **Altera max V 5M570ZF256C5N** development kit along with an **MCP4921 12-bit SPI DAC** to observe the analog data in an oscilloscope.

The problem with the MCP4921 DAC is that its settling time is 4.5 μs, which is around 200 kHz. So it is pointless to send data at 1 MHz if the DAC will not be able to settle to that analog value within the timeframe. Therefore, I had prescaled the clock frequency to run at 100 Khz, and the converter switching frequency at 10 Khz, these are definitely scalable since FPGA can easily run it at high frequencies, but due to hardware constraints, I was not able to properly simulate the frequencies, but the waveforms are captured accurately to its shape.

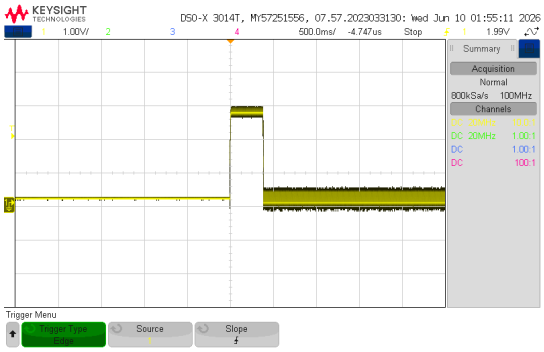


Fig. 28. Inductor Current full waveform

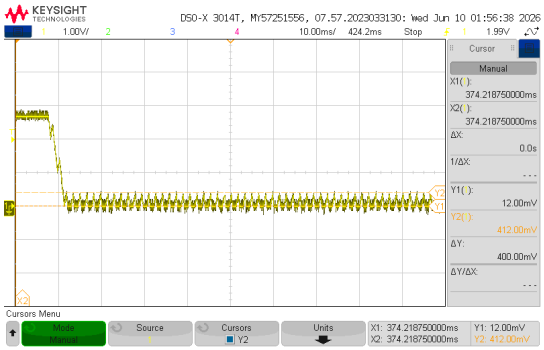


Fig. 29.

1) 1. Fixed-Point to DAC Data: $\mathrm{D_{val}} = \frac{I_L \times 2^{16}}{2^8} = I_L \times 256$

1) 2. DAC Hardware Formula: $\mathrm{V_{probe}} = \frac{D_{\mathrm{val}}}{4096} \times V_{\mathrm{ref}}$

1) 3. Combined Equation: $\mathrm{V_{probe}} = \frac{I_L \times 256}{4096} \times V_{\mathrm{ref}} = \frac{I_L \times V_{\mathrm{ref}}}{16}$

1) 4. Final Conversion: $\mathrm{I}_L = \left(\frac{16}{V_{\mathrm{ref}}}\right) \times V_{\mathrm{probe}}$

Since Vref = 3.3V, Amps = 4.848 * V probe. From the above figure, V probe = 400mV, which translates to around 2A, which is exactly what is observed in PLECS and ModelSim simulations.

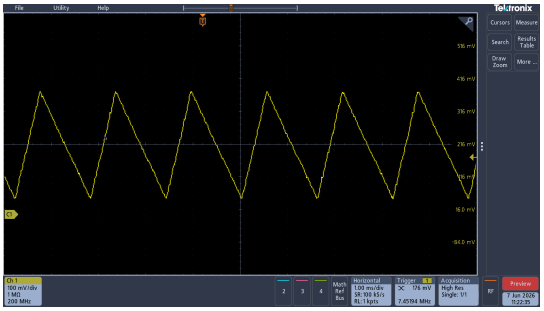


Fig. 30. Triangular Inductor Current Waveform

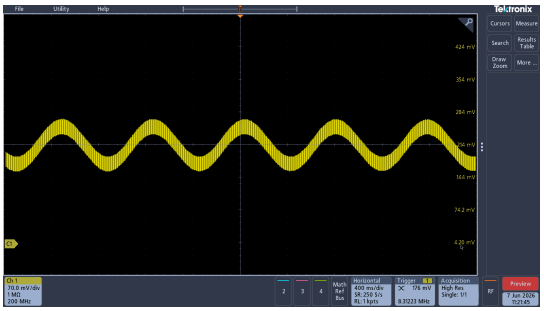


Fig. 31. Low Frequency Oscillations observed in boost converter

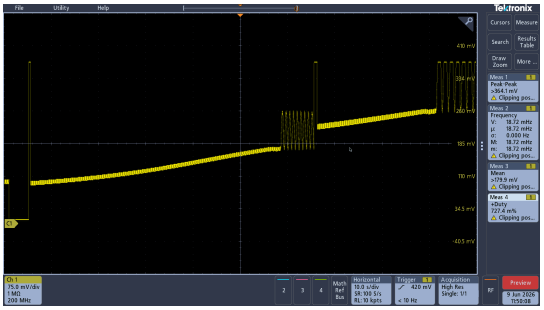


Fig. 32. Inductor Current variation durind duty variation from 0.334 to 0.6

The slope or increase in inductor current is around 20 mV, that implies a slow increase in average inductor value, this is actually observed in PLECS simulation as well, since right after the transient, inductor current is in DCM, then slowly before the low frequency oscillations kick in, it moves into CCM, this obviously implies the average inductor current has increased.

I had changed the duty to 0.6 right after the low frequency oscillations kicked in, so a spike in inductor current after the oscillations are linked to duty change.

VIII. Conclusion

The goal of this work was to develop a practical, low-cost real-time HIL simulator that bridges the gap between high-end commercial systems and budget-constrained developers. By designing a custom interface card and implementing a dual-stage emulation strategy—starting with an averaged model on a C2000 DSP and progressing to a switching model on an FPGA—we demonstrated that accessible embedded hardware can successfully handle real-time plant simulation. The hardware experiments confirm this approach, capturing both steady-state dynamics and transient oscillations accurately, proving that engineers can reliably verify control loops without relying on expensive proprietary platforms.

IX. Future Work

Our next steps will focus on expanding the platform's performance and versatility. First, we plan to migrate the switching models to a higher-performance FPGA architecture to eliminate the frequency limitations imposed by our current hardware prototype. Second, we aim to extend the system's codebase beyond the boost converter to support a broader range of topologies, including buck-boost, flyback, and bidirectional dual-active-bridge converters, ultimately creating a versatile, open-source library for power electronics research.

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